

Asking Testable Analytical Questions

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Why are we still talking about the “question”?

The Streetlight Effect

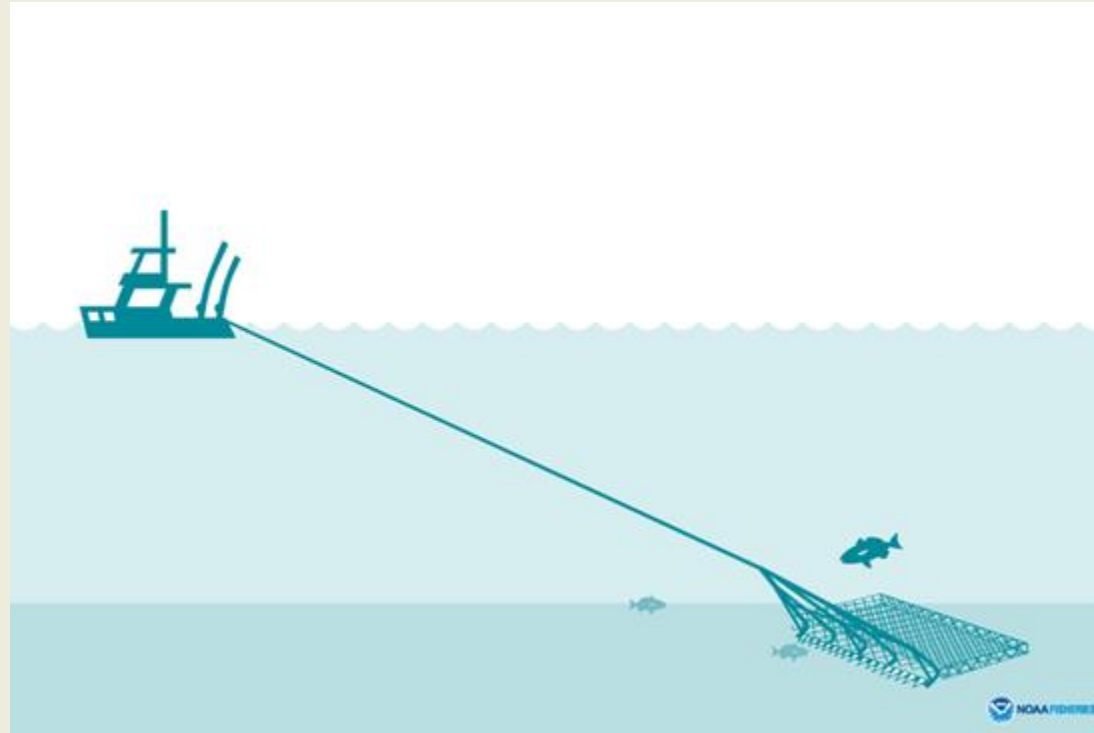


The Streetlight Effect

When you start with the data, you may:

- Gravitate toward variables that are already collected.
- Miss important factors outside the dataset.
- Produce technically correct analysis that does not address actual performance problems.

Data Dredging



probing data using unplanned analyses and then reporting salient results without accurately describing the processes by which the results were generated. (Erasmus, Holman & Ioannidis , 2022)

Data Dredging

- Most often done by repeated statistical testing
- Can produce misleading or false conclusions

“We have GPS data. What relationships can we find?”

- If enough relationships are tested, some are likely to appear statistically significant by chance.

Available Data Can Rewrite Questions

Original concern

- Why does the team struggle to create good shots against zone defense?

Available spreadsheet

- Points, rebounds, assists, and minutes

Change the question

- Who scores the fewest points?

Collect more data

- Play-by-play game data

The replacement is answerable, but it **does not answer** the original concern.

Building a Question That Can Guide the Analysis

Analysis must start with clarity

“Our team fades late in games”

Clarifying questions:

- What does **fades** mean?
- Which team, players, games, or situations?
- Compared with what?
- What evidence would support or challenge the belief?
- What decision would change if the belief were true?

The starting point must be observable and the purpose must be clear.

Good Questions Connect Curiosity to a Decision

A useful analytical question identifies:

1. **The decision or purpose** : Why do we need the answer?
2. **The outcome** : What are we trying to understand?
3. **The comparison** : Relative to whom, what, or when?
4. **The context** : Under which conditions?
5. **The evidence** : What observations could answer it?

The question is the bridge between the sport problem and the analysis.

Begin With the **Decision**

“I want to know whether our players are tired.”

Possible decisions include:

- Should today's practice be modified?
- Should a player's minutes be restricted tomorrow?
- Should the weekly training plan change?
- Should the team adopt a new recovery strategy?

The same concern can lead to different questions because the decisions are different.

Define the **Outcome** Before Choosing the Metric

How do we want to measure how “tired” an athlete is?

- “tired” could refer to *fatigue*, which could mean:
 - Reduced physical performance
 - Increased perceived effort
 - Muscle soreness
 - Lower motivation or alertness

A metric represents one part of a concept; it is not automatically the concept itself.

Comparisons Give Numbers Meaning

“Did the athlete complete a high workload?”

High compared with:

- The athlete's usual workload?
- The planned workload?
- Teammates at the same position?
- Competition demands?
- A threshold associated with a particular goal?

Without a reference point, “high,” “low,” “good,” and “poor” are usually empty labels.

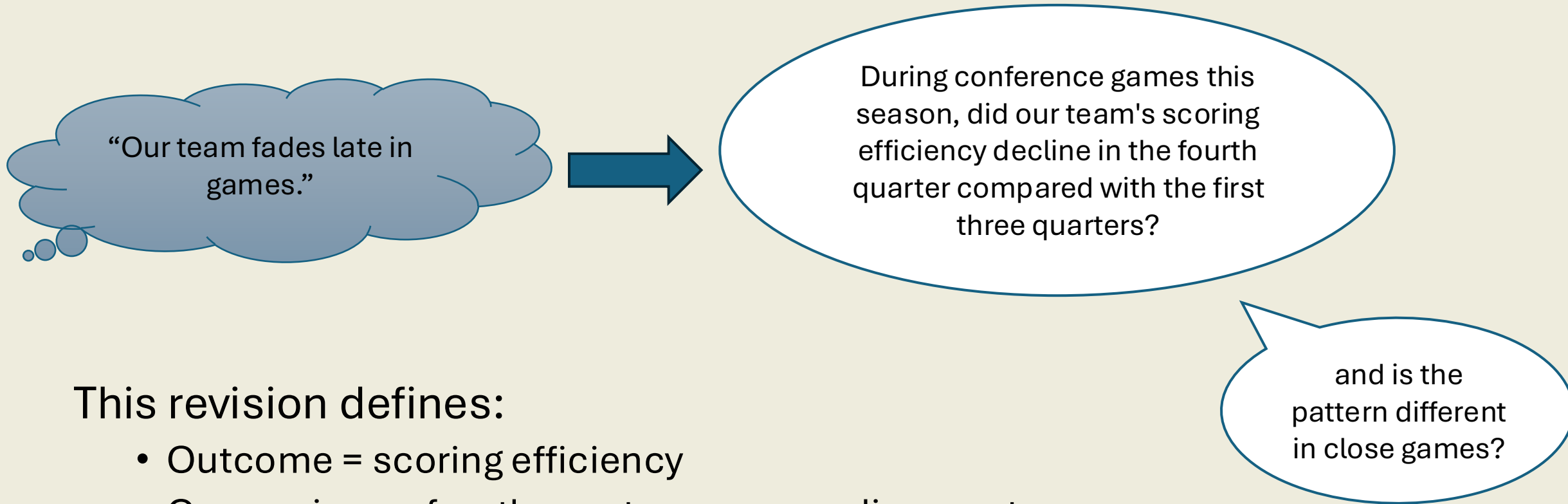
Context Determines What the Comparison Means

The same performance value may mean something different depending on:

- Position or role
- Opponent quality
- Game state
- Training phase
- Injury or return-to-play status
- Age and competition level

Useful questions define just enough context to make the answer interpretable.

From Concern to Question

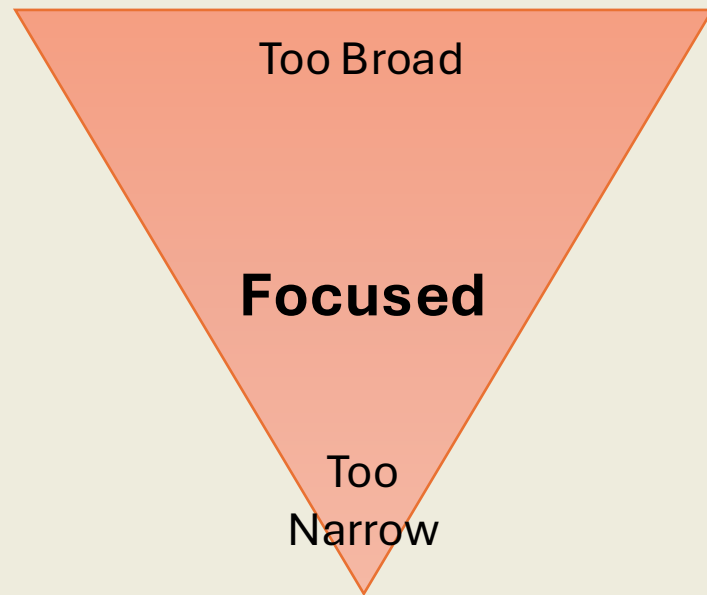


This revision defines:

- Outcome = scoring efficiency
- Comparison = fourth quarter versus earlier quarters
- Context = conference games

What else needs to be resolved?

Precision Does Not Mean Small Questions

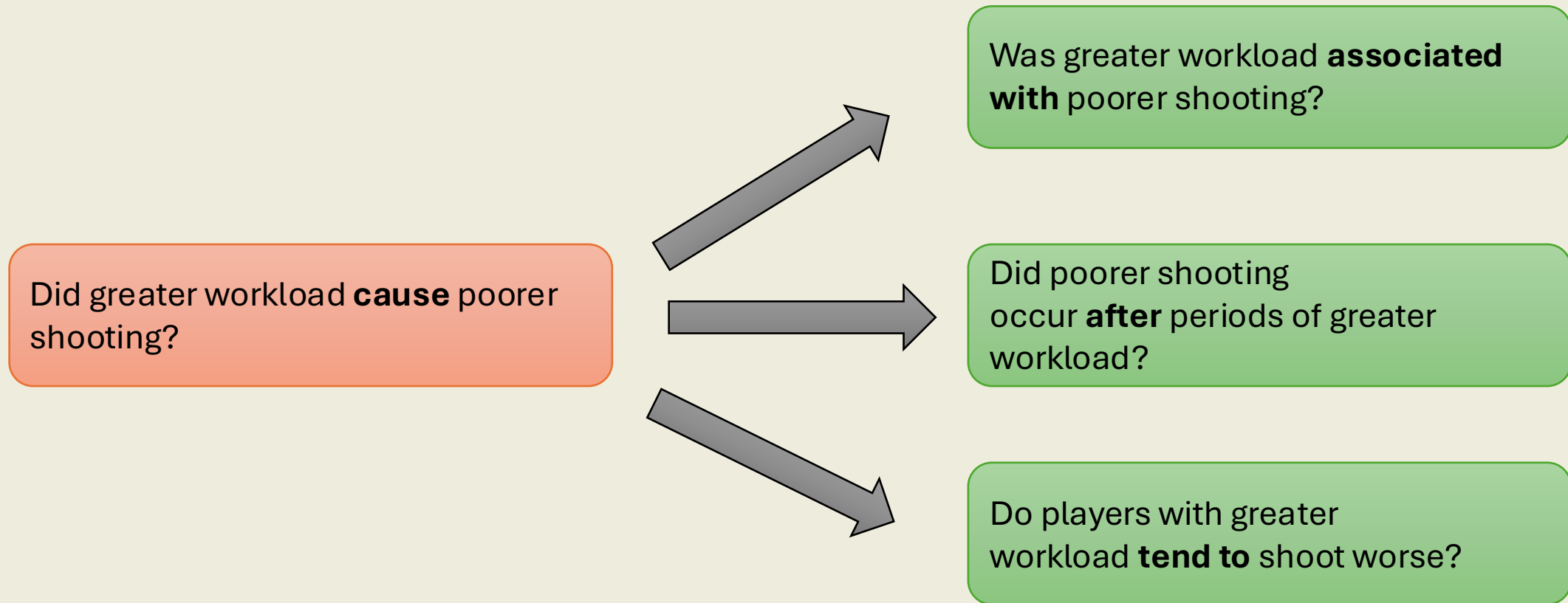


- What makes a basketball team successful?
- How does turnover rate relate to offensive efficiency across conference games?
- Did the turnover with 72 seconds remaining in Q3 cause Tuesday's loss?

A useful scope is narrow enough to **investigate reasonably** and broad enough to **matter**.

Warning on Causal Language

The wording of the question should match what the evidence can support.



A Question Can Be Useful Even If the Answer Is “We Cannot Tell”

Sometimes the conclusion of an investigation may be:

- The required variable was not measured.
- The comparison group is inappropriate.
- The sample is too limited.
- The measurement does not represent the intended concept.
- The design cannot support the claimed explanation.

Recognizing limits is a crucial part of analytics

Data Availability Matters but It Should Not Lead

Practical Steps:

1. **Define** the sport **concern** and **decision**.
2. Write the **ideal question**.
3. **Identify the evidence** the question requires.
4. **Inspect** what data are actually available.
5. Revise the **scope** without changing the question's purpose.
6. State what the **available data cannot establish**.

Step 5 should refine the question, not replace it.

Exploring the data for patterns

- **Exploratory analysis** is a legitimate analytical approach used to identify patterns, relationships, and potential questions.
- Findings discovered through exploration should **tested with new data** before making stronger explanatory, predictive, or prescriptive claims.

CS1: Establish the Project Direction

In this you will identify:

1. A sport and domain that interests you
2. A real observation, concern, or decision
3. Who would use the answer
4. An initial analytical question
5. The ideal evidence needed
6. Data sources that may be feasible

Keep the concern (step 2) and the question (step 4) separate

Summary & Looking Ahead

- Start with a sport concern or decision.
- Define the outcome, comparison, and context.
- Use causal language only when the evidence supports it.
- Let feasibility refine the scope without replacing the purpose.
- Treat limitations as part of the answer.
- Next: at-home case study (generate question and identify dataset)